

Uncovering the Limits of Spatial Reasoning in Vision-Language Models

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Abstract

Vision-language models are increasingly used for tasks that require understanding the physical world, yet they consistently struggle with spatial reasoning, and it is still unclear which internal parts are responsible. This matters because it limits how reliably these models can be used in real-world applications that depend on seeing and understanding a scene. We evaluate the Qwen3-VL-30B model on the LEGO-Lite benchmark, which tests four spatial reasoning categories: height, position, rotation, and ordering. To find the source of its errors, we tracked the model’s internal expert routing across all 48 layers for 400 questions. This showed that image and text inputs activate largely separate sets of experts, with image processing spreading across more experts than text. We then ranked layers by a z-score and ran ablation experiments, disabling the worst-scoring layers during inference. Disabling the MoE sublayer in the three worst layers raised accuracy from 23.0% to 26.5% on the test set, compared to a 23.2% average when random layers were removed instead. However, McNemar’s test gave $p=0.25$, and a held-out validation set showed accuracy actually dropped from 29.0% to 27.0% under the same setup, so the test-set improvement did not hold up. These results suggest that this model’s spatial reasoning failures are spread broadly across layers rather than concentrated in a few, and that targeted layer ablation alone is not enough to produce a real improvement.

1 Introduction

Vision-language models built on a Mixture-of-Experts (MoE) backbone only activate a small fraction of their parameters per token, sending each token to a handful of specialist subnetworks (“experts”) chosen from many available at each layer (Shazeer et al., 2017; Fedus et al., 2022; Jiang et al., 2024). This sparsity is usually there for efficiency, but it also gives the model something a dense transformer does not have: at every layer, every token leaves an explicit, inspectable record of which internal specialists handled it. That record is a direct handle for interpretability work, since it lets us ask not just *whether* a model fails at a task, but *which specific internal components were involved* when it did.

We study this question on a concrete, currently-hard task: multi-step spatial reasoning over LEGO assembly images, using the LEGO-Puzzles benchmark (Tang et al., 2025) and its four-category LEGOLite subset (height, position, rotation, and ordering questions). Vision-language models are known to do poorly on this benchmark compared to general visual question answering, and the model we study, Qwen3-VL-30B-A3B-Instruct (Qwen Team, 2025), is no exception. Across our experiments it answers well under a third of LEGOLite questions correctly, only a bit above the four-way random-guess baseline of 25%. This is not just a benchmark curiosity: autonomous and multi-agent systems that plan physical actions from visual input depend on exactly this kind of multi-step spatial reasoning to interact reliably with an environment. A model that fails to track height, position, rotation, or assembly order is a concrete bottleneck for deploying these systems in the real world, not only for scoring well on a puzzle benchmark.

Our starting hypothesis was simple: if the model’s spatial reasoning failures are concentrated in a small number of layers or experts, whose routing behavior looks measurably different on questions it gets right versus wrong, then finding and disabling those specific components should recover some accuracy. This hypothesis is appealing because it is direct. If true, this would prove that spatial reasoning failure in this model is a localized and fixable problem rather than a diffuse one. It would also yield a concrete method for improving accuracy on a hard benchmark without retraining by systematically scoring experts, clustering the worst performers, and disabling them.

We built a four-phase pipeline to test this. Phase 0 checks where the model sits accuracy-wise against comparison models on LEGOLite, confirming it is a reasonable subject for this kind of study. Phase 1 uses Google Colab to check whether the model routes image and text tokens through distinguishable sets of experts, which is a necessary precondition for the later analysis. Phase 2 scales expert-activation logging up to the full 400-question LEGOLite set on a cloud GPU, producing the dataset used by every later phase. Phase 3 is the core of this paper. We rank all 48 decoder layers by a z-score that measures how much more often each layer’s experts fire on wrong answers than the model’s overall error rate would predict. We then cluster the experts within the worst-scoring layers by their activation fingerprints, and run a controlled ablation study comparing three types of intervention: skipping the whole decoder block, skipping only the MoE sublayer, or zeroing only the flagged experts’ gate weights. Each is tested against a baseline and a random-ablation control, with significance assessed by McNemar’s test and generalization checked on a held-out question set that scoring and clustering never touched.

The result of Phase 3 is negative, and we report it as such. No ablation configuration tested produces a statistically significant change in accuracy, and targeted ablation of the layers our method flagged as ”worst” does not beat ablating the same number of randomly chosen layers. The single best-looking configuration on our 200-question test set, MoE-only ablation of the three worst layers, gains 3.5 percentage points over baseline. But that gain is not significant by McNemar’s test ($\chi^2 = 1.33$, $p = 0.25$). It also fails to hold up on a held-out set of 200 questions that were never used for scoring or clustering, accuracy there *drops* by 2 points instead. We treat this outcome carefully. Since the z-scores did not exceed a magnitude of 0.055 and not a single expert clustering reached a silhouette score above 0.25, we concluded that our underlying signal was weak. Due to this, it was difficult to say whether spatial reasoning failure is genuinely not localized to any small set of experts, or whether it is localized but our scoring method was too coarse to find it. Section 8 explains why we think this ambiguity itself is worth reporting, and Section 10 proposes a causal-tracing experiment in which activation patching between a clean and a corrupted visual input serves as a follow-up that directly tests the same question rather than relying on a correlational proxy.

2 Related Work

Mixture-of-Experts models. Sparsely-gated MoE layers route each token to a small subset of expert subnetworks chosen by a learned gating function, letting model capacity scale largely independent of per-token compute cost (Shazeer et al., 2017). Switch Transformers (Fedus et al., 2022) simplified routing down to a single top-1 expert per token at trillion-parameter scale, and Mixtral (Jiang et al., 2024) showed that top-2 MoE routing in a smaller open model can match much larger dense models at a fraction of the active-parameter cost. Zoph et al. (2022) studied design choices in sparse expert models, including routing algorithm and load balancing. However, interpretability work asking whether a specific expert matters for a specific failure mode remains fairly rare, which is part of what motivates this study. Our finding, that expert-level scoring only surfaced weak, non-generalizing signal, agrees with Wang et al. (2026), who analyzed routing across several MoE language models and found a domain-invariant “standing committee” of experts that dominates routing regardless of task or domain, concluding that semantic specialization in MoE routing is far less common than usually assumed. Our result can be read as an instance of this same pattern in a

vision-language MoE model.

Spatial reasoning benchmarks for vision-language models. LEGO-Puzzles (Tang et al., 2025) tests multi-step spatial reasoning through LEGO assembly tasks covering spatial understanding, single-step sequential reasoning, and multi-step sequential reasoning. It finds that current multimodal LLMs, while decent at simpler visual question answering, fall short of human performance on these tasks, especially the multi-step and generative ones. This is not an isolated finding: Kamath et al. (2023) show that across 18 vision-language models even simple left/right spatial relations are handled unreliably, tracing part of the problem to a lack of reliable spatial supervision in common pretraining data. Spatial reasoning weakness in vision-language models therefore looks like a general phenomenon rather than a quirk of one benchmark or model, which is part of why we treat Qwen3-VL-30B-A3B’s failures on LEGOLite as worth explaining mechanistically instead of dismissing as benchmark-specific noise. We use the LEGOLite subset (height, position, rotation, ordering) throughout this work, since it isolates four distinct categories of spatial relation while keeping evaluation cost manageable across our multi-phase pipeline.

Causal interpretability methods. Activation patching, also called causal tracing, intervenes on a model’s internal activations at a specific layer and token position while holding the rest of the computation fixed, to test whether that location is causally responsible for a specific output (Meng et al., 2022; Vig et al., 2020). This differs from our Phase 3 approach, which uses correlational routing statistics, and how often an expert fires on wrong answers to pick ablation targets rather than directly testing causal responsibility at every layer. Section 10 proposes adapting causal tracing to our setting as a more direct follow-up.

Layer removal and depth redundancy. Gromov et al. (2024) show that a large fraction of a dense LLM’s deeper layers can be removed with only limited performance loss, which matters directly for our random-ablation control: if a model tolerates arbitrary layer removal fairly well in general, a targeted ablation has to be compared against a random one before its effect can be credited to the specific layers chosen, rather than to depth removal in general.

3 Model and Benchmark

Model. Qwen3-VL-30B-A3B-Instruct is a vision-language MoE model with 48 decoder layers, each with a mixture-of-experts sublayer of 128 experts, of which the router picks the top 8 per token (`num_experts_per_tok = 8`, `decoder_sparse_step = 1`), meaning every layer, not just alternating ones, has an MoE block. Routing is done by a `Qwen3VLMoeTextTopKRouter` module, accessible at `layer.mlp.gate` in the Hugging Face implementation. Its forward pass returns per-token routing logits, post-softmax routing weights, and the selected expert indices, and we hook it directly to log routing behavior.

Benchmark. LEGOLite is the four-category subset of LEGO-Puzzles (Tang et al., 2025) we use throughout: 400 questions split evenly across height (spatial understanding via height comparison), position (relative spatial positioning), rotation (single-step rotation reasoning), and ordering (multi-step sequential assembly ordering), 100 questions per category. Height, position, and rotation are standard multiple-choice questions with up to five lettered options (A through E). Ordering questions work differently: each lettered option (usually A through D) is itself a single candidate assembly-step image, and the model must output the full letter sequence (e.g. “BDAC”) that correctly orders those steps. This is graded by the official LEGO-Puzzles code as an exact match against the full ground-truth sequence, so chance performance for ordering is roughly 1-in-24 (for 4 steps) rather than 1-in-4.

A scoring difference between Phase 0 and Phases 1 through 3. Phase 0 (Section 4) uses the official LEGO-Puzzles generation-and-exact-match grading described above. Phases 1 through 3, by contrast, hook the model’s internal routing gates, which needs a single constrained forward pass rather than free-form multi-token generation. Those phases score every category, including ordering, by comparing the model’s single predicted token against only the *first* letter of the ground-truth sequence for ordering questions (and the full answer letter for the other three categories). This is a coarser proxy specifically for ordering: it only asks whether the model gets the first assembly step right, not whether it produces the whole correct order, and its chance baseline is close to the same 4-choice 25% as the other categories rather than the roughly 4% chance level of the official full-sequence task. We flag this because it means the “ordering” accuracy numbers in **Table 2** (Phase 0) and **Tables 3** onward (Phases 2 and 3) are not measuring exactly the same thing, and should not be read as a direct before-and-after comparison for that category. **Table 1** summarizes the distinction.

	Phase 0	Phases 1–3
Grading method	Full-sequence exact match	First-letter-only match
What is checked	Entire 4-step order correct	Only the first assembly step correct
Chance baseline	~4% (1-in-24)	~25% (1-in-4)

Table 1: Ordering-category grading differs between Phase 0 and Phases 1 through 3; the two are not directly comparable.

4 Phase 0: Baseline Accuracy Benchmarking

Before instrumenting the model internally, we first checked where Qwen3-VL-30B-A3B-Instruct sits accuracy-wise on LEGOLite compared to other vision-language models, using the Fireworks AI serving API (no local GPU or routing access needed for this phase). This baseline motivates the rest of the paper: a model that had already solved the benchmark would give interpretability work nothing to explain, and a model at chance on every category would leave no structure in its correct answers to compare against its incorrect ones.

Model	Overall	Height	Position	Rotation	Ordering
Qwen3-VL-30B-A3B (Fireworks)	31%	34%	42%	34%	14%
Gemini-2.0-Flash	44.25%	35%	47%	49%	46%
GPT-4o Mini	13.75%	29%	10%	12%	4%

Table 2: Phase 0 baseline accuracy by category on the same 400-question LEGOLite subset used throughout this paper. Qwen3-VL-30B-A3B was run via the Fireworks AI API; Gemini-2.0-Flash and GPT-4o Mini figures are the LEGOLite-only subset of their full LEGO-Puzzles evaluation.

Qwen3-VL-30B-A3B lands between the two comparison models overall, clearly ahead of GPT-4o Mini and clearly behind Gemini-2.0-Flash. That is a reasonable middle ground for an interpretability case study: the model is capable enough that its correct answers are not accidental, but weak enough that its failures are frequent enough to study in bulk. The ordering category is the clear outlier at 14% accuracy, well below height, position, and rotation (34–42%), and below the roughly 25% chance level the other three categories are graded against. This number was originally logged as 0%, but that turned out to be a grading-pipeline bug rather than a real result. The evaluation harness’s answer-matching step failed to even attempt a real comparison on any of the 100 ordering questions in this run, including rows where the model’s raw

prediction was a character-for-character match with the ground-truth sequence. Grading those same 100 questions by simple exact-string match, the same standard already used successfully for height, position, and rotation, gives 14% (14/100). The same 100 ordering questions graded normally for Gemini-2.0-Flash in the same harness (46%, **Table 2**), which confirms this is a bug specific to how the grader parsed Qwen3-VL-30B-A3B’s responses in this run, not a flaw in the ordering category or benchmark itself. Ordering remains the model’s weakest category by a wide margin, but not the complete, zero-correct failure originally reported. This category-level gap is the first hint, before any routing analysis, that the model’s spatial reasoning failures might not be uniform across task types. We return to per-category differences throughout Phases 1 through 3.

5 Phase 1: Modality-Level Routing Analysis

Before asking whether specific experts are responsible for spatial reasoning errors, we first checked a precondition: does the model even route image tokens and text tokens through different sets of experts? If image and text tokens were routed almost identically, expert-level analysis of visual reasoning would tell us nothing, since there would be no dedicated visual processing pathway to trace failures back to.

Running on a single Google Colab GPU, we logged full router probabilities for all 48 MoE layers on one representative LEGOLite sequence (2,768 tokens total: 2,754 image tokens and 14 text tokens). We then compared the routing distribution over image-token positions against the distribution over text-token positions. One scope limitation up front: this analysis covers one sequence rather than an aggregate over all 400 questions, and it logs the top-4 activated experts per token rather than the top-8 used in Phases 2 and 3 (an earlier, smaller activation count used only in this exploratory pass). We report it as a directional precondition check, not a precisely quantified population-level estimate.

As a result, image and text tokens route through measurably different, and largely disjoint, sets of experts. The Jensen-Shannon divergence between the image-token and text-token routing distributions is 0.0215 ($\text{KL}(\text{image} \parallel \text{text}) = 0.0213$, $\text{KL}(\text{text} \parallel \text{image}) = 0.0217$). Of the top 10 most-activated experts under each token type, *zero* are shared: all 10 image-preferred experts and all 10 text-preferred experts are mutually exclusive. Routing entropy, averaged across all 48 layers, is also higher for image tokens (6.954 bits) than for text tokens (6.589 bits), meaning text-token routing concentrates on a narrower, more repeatable set of preferred experts, while image-token routing spreads more evenly across the expert pool. **Figure 2** shows this gap holds at nearly every individual layer, not just on average.

Global Routing Heatmaps

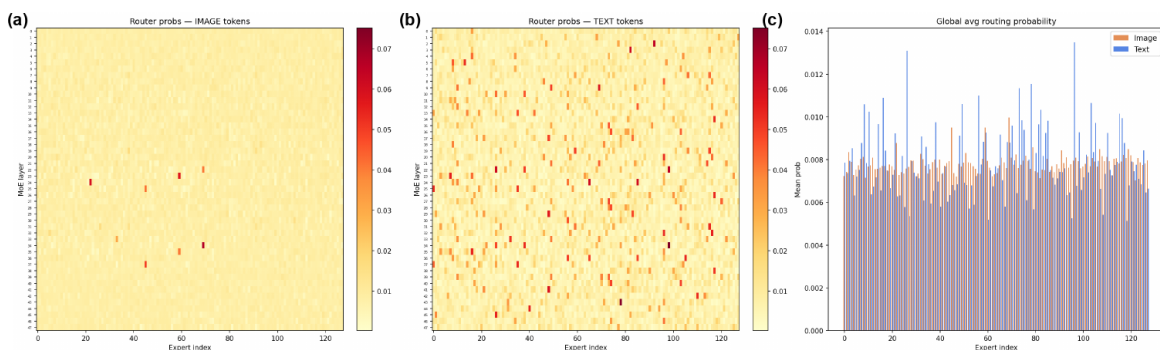


Figure 1: Router probabilities by MoE layer and expert index for image tokens (a) vs. text tokens (b) on the representative sequence, and each expert’s global mean routing probability under image tokens vs. text tokens (c). Text-token routing (c, blue) shows sharper, higher-magnitude spikes on a smaller set of experts than image-token routing (orange), consistent with the lower text-token entropy reported above.

Routing Entropy per Layer

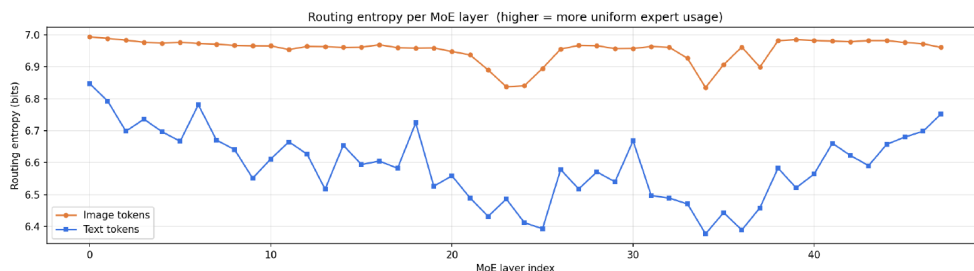


Figure 2: Routing entropy per MoE layer for image tokens vs. text tokens. Image-token routing (orange) is more uniform across experts, higher entropy, than text-token routing (blue) at nearly every one of the 48 layers, not just in the layer-averaged figure reported in the text.

This is a necessary, though not sufficient, precondition for the expert-level localization hypothesis tested in Phase 3: if visual and textual processing were not separated at the routing level, there would be little reason to expect visual reasoning failures to trace back to a specific subset of experts at all. Given the single-sequence scope noted above, we treat this as supportive evidence for that precondition, not a fully quantified claim about the model’s behavior across the whole benchmark.

6 Phase 2: Expert Activation Logging at Scale

Phase 2 moved from Colab to a cloud A100 80GB GPU (RunPod) to log full routing behavior, all 48 layers $\times$ 128 experts, over all 400 LEGOLite questions in a single inference pass, using 4-bit NF4 quantization to fit the 30B-parameter model on one GPU. We ran this pipeline twice, producing two independent activation logs (Run 1 and Run 2), so we could sanity-check the stability of accuracy and routing statistics before settling on one dataset for the heavier Phase 3 analysis.

Run	Overall	Height	Position	Rotation	Ordering
Run 1	24.25%	32%	22%	23%	20%
Run 2	25.25%	32%	25%	24%	20%

Table 3: Phase 2 accuracy by category, two independent runs on the same 400 LEGOLite questions.

The two runs agree closely (within 1 percentage point overall, and within 1 to 3 points per category), which we take as evidence that the quantized model’s behavior is stable enough for Run 2’s activation log to serve as a reliable source dataset for the layer-scoring and clustering analysis in Phase 3. We use `data/phase2/runpod_second/results.json`, per-question routing weights and correctness labels for all 400 questions, as that source dataset from here on.

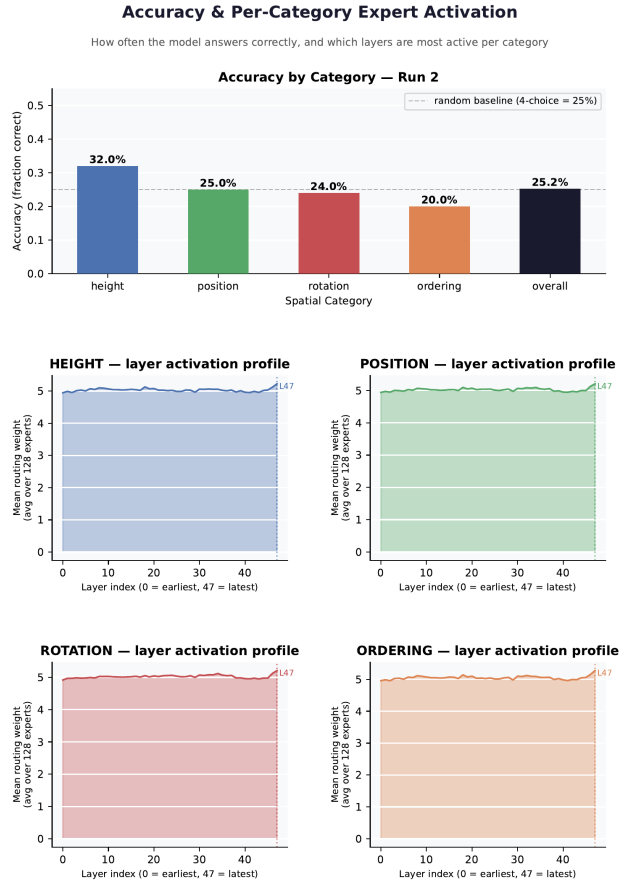


Figure 3: Run 2 accuracy by category alongside each category’s mean routing weight by layer, averaged over all 128 experts and all questions in that category. Every category’s activation profile is nearly flat across all 48 layers, which foreshadows the weak, non-localized layer signal we find later in Phase 3.

Visual Token Expert Activation Heatmaps — Run 2
(Layer × Top-32 most active experts; colour = mean routing weight)

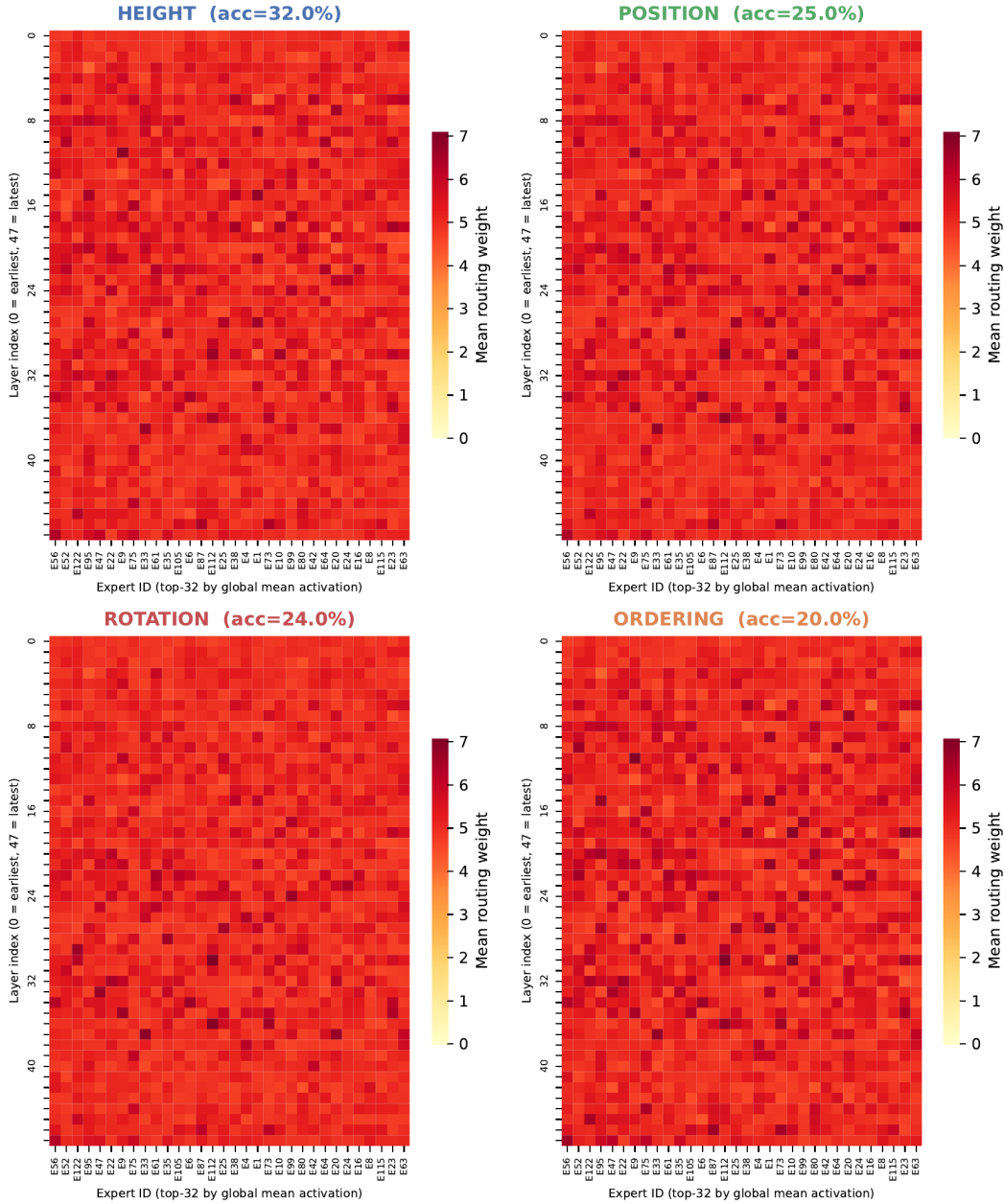


Figure 4: Run 2 visual-token expert activation, layer by top-32 globally most active experts, one heatmap per category. The four categories share a broadly similar activation pattern across experts and layers despite their differing accuracy, consistent with the flat per-layer profiles in **Figure 3**.

We also checked whether experts are generic or category-specific by taking each category’s top 20 most-activated experts and counting how many categories’ top-20 lists each expert appears in. Across the four categories, 34 distinct experts appear in at least one top-20 list. Of these, 10 appear in only one category’s list (category-specific), 8 appear in all four (generic, general-purpose experts used regardless of question type), and the rest appear in two to three. So both kinds of experts exist side by side, some genuinely

specialize to one category, but a similarly sized group is used across all four, which is consistent with the flat, similar-looking activation patterns in **Figure 4**.

7 Phase 3: Statistical Localization and Causal Ablation

Phase 3 is the central experiment of this paper. It has six steps: split the Phase 2 data into a test and a holdout set, score all 48 layers for their correlation with wrong answers, build expert activation fingerprints for the worst-scoring layers, cluster those experts to isolate a candidate “bad” subset, ablate the candidates under three intervention types with a random-ablation control, and finally validate the best configuration on the held-out set.

7.1 Data Split

We split the 400 Phase 2 questions into a 200-question test set and a 200-question holdout set, stratified by category so each set has roughly 50 questions per category. All layer scoring and expert clustering (Steps 2 through 4 below) uses only the test set. The holdout set stays untouched until the final validation step, so whatever configuration we settle on gets one genuine chance to generalize, rather than being tuned against the same data it is validated on. On the test set, the model’s baseline error rate is 76.5% (23.5% baseline accuracy), which becomes the null-hypothesis wrong-answer rate for the z-score computation below.

7.2 Layer Scoring by Z-Score

For each expert e in each layer, we binarize its routing decisions (the router’s top-8 selection is equivalent to a binary indicator per question, since softmax preserves rank order), then compute a z-score comparing its observed wrong-answer rate to what the model’s overall error rate would predict by chance:

$$z_e = \frac{\text{observed_wrong}_e - n_e \cdot \bar{p}}{\sqrt{n_e \cdot \bar{p}(1 - \bar{p})}} \quad (1)$$

Here n_e is the number of test questions expert e was selected on, $\bar{p} = 0.765$ is the base error rate, and observed_wrong_e is how many of those n_e questions the model got wrong. This formula automatically discounts experts with few activations, since the denominator scales with $\sqrt{n_e}$: an expert selected on only two or three questions cannot produce a large z-score regardless of outcome, with no separate frequency-weighting step needed. A layer’s score is the average z_e across its 128 experts.

Layer	Average z-score
41	+0.0546
42	+0.0406
43	+0.0368
19	+0.0360
7	+0.0302

Table 4: Top five worst-scoring layers by average expert z-score, out of 48 layers scored. Maximum possible signal strength in this dataset was weak: no layer’s z-score magnitude exceeded 0.055.

We flag these five layers as candidates for the clustering and ablation steps that follow, ranked from worst (layer 41) to fifth-worst (layer 7). We note up front, and return to this in Section 8, that a z-score of 0.05 is very weak evidence next to the $z > 2$ threshold conventionally used to flag a single outlier expert. No individual expert in any of the 48 layers exceeded $z = 2$.

7.3 Expert Clustering

For each of the five candidate layers, we built a 128×200 binary activation fingerprint matrix, one row per expert, one column per test question, with a 1 if that expert was selected for that question. After dropping experts with no activations across the 200 questions (all-zero fingerprints, which Jaccard distance below cannot meaningfully compare), we computed pairwise Jaccard distance between the remaining experts,

$$d_J(A, B) = 1 - \frac{|A \cap B|}{|A \cup B|}, \quad (2)$$

where A and B are the sets of questions each expert activates on. Jaccard distance fits here because each fingerprint is roughly 93.75% zeros (8 of 128 experts selected per question). Therefore, a metric that counts shared zeros as similarity, like Euclidean distance, would judge almost all expert pairs as similar, regardless of their actual activation overlap.

We ran k-medoids clustering with $k \in \{2, 4, 6, 8\}$ on each layer’s Jaccard distance matrix, and kept the k with the highest silhouette score. A cluster’s wrong-answer rate is the fraction of (expert, question) activation pairs in that cluster where the question was answered incorrectly; a cluster is flagged “bad” if this rate is above the 76.5% base error rate.

Layer	Chosen k	Best silhouette	Bad-cluster experts flagged
41	8	0.147	24
42	4	0.236	14
43	2	0.069	4
19	2	0.073	1
7	2	0.101	12

Table 5: Clustering results per candidate layer. Silhouette scores below roughly 0.25 to 0.5 indicate weak or overlapping cluster structure; none of our five layers reached even the low end of that range.

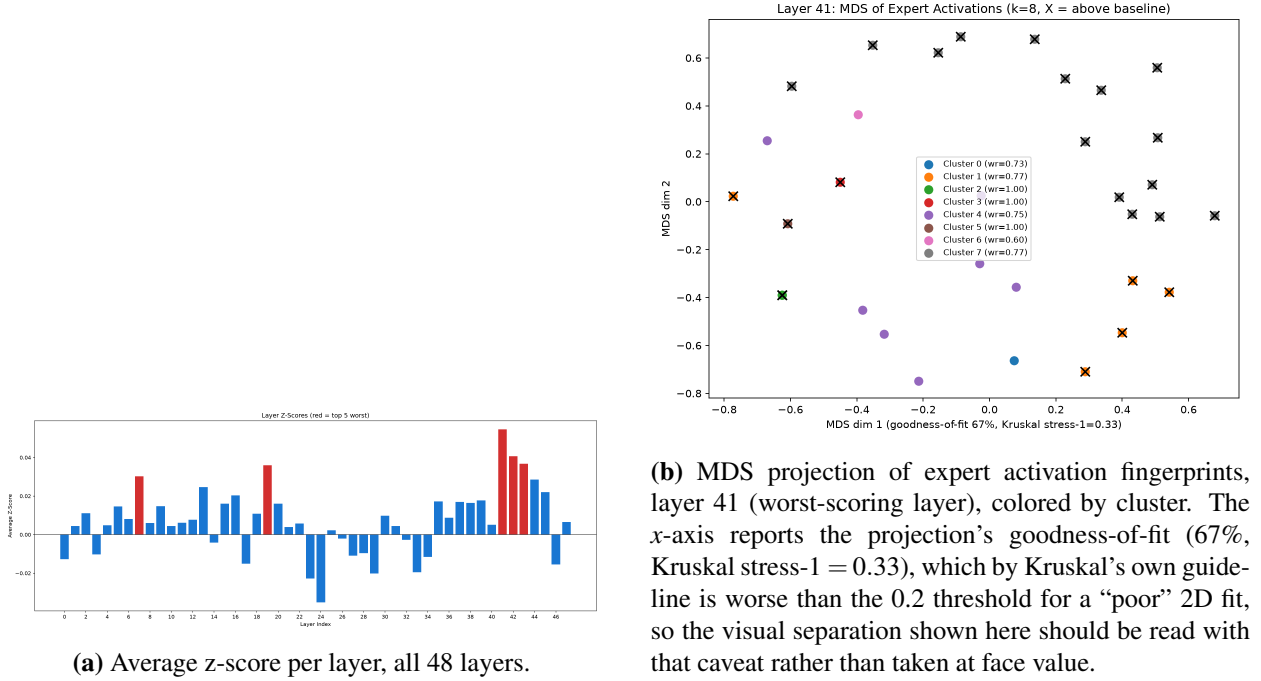


Figure 5: Layer scoring and clustering diagnostics. Even the worst-scoring layer’s expert clusters show only modest visual separation in the 2D MDS projection, consistent with the weak silhouette scores in [Table 5](#) and the weak 2D goodness-of-fit reported in the MDS panel.

7.4 Ablation Design

We test three ways of disabling a decoder layer’s contribution, isolating which component of the layer, if any, is responsible for the flagged behavior:

- **Full block ablation:** the entire decoder layer (attention and MoE) is skipped; the residual stream passes through unchanged, as if the layer did not exist.
- **MoE-only ablation:** only the MoE sublayer’s output is zeroed; the attention sublayer still runs normally. This isolates whether the routed experts specifically, rather than the layer’s attention computation, are responsible for degraded accuracy.
- **Expert-level ablation:** only the gate weights for the flagged “bad” cluster experts are zeroed and the remaining experts’ weights renormalized to sum to 1, leaving the rest of the layer, including its other, non-flagged experts, untouched.

Part A compares all three intervention types on the single worst layer (layer 41) against baseline. **Part B** takes whichever intervention type performed best in Part A (MoE-only, in our results) and scales it to the 1, 3, and 5 worst layers. **Part C** is the random-ablation control: for each layer count (1, 3, 5), we run 10 trials ablating that many randomly chosen layers with MoE-only ablation, giving a null distribution to compare targeted ablation against.

We test the significance of each ablation run against baseline with McNemar’s test on paired per-question outcomes, and test whether targeted ablation beats random ablation with a z-score against the random trials’ mean and standard deviation. All reported accuracies include 95% confidence intervals, $p \pm 1.96\sqrt{p(1-p)/n}$.

7.5 Results

Run	Accuracy	95% CI	McNemar p
Baseline (no ablation)	23.0%	[17.2%, 28.8%]	n/a
Full block ablation	22.0%	[16.3%, 27.7%]	0.850
MoE-only ablation	24.0%	[18.1%, 29.9%]	0.831
Expert-level ablation	24.0%	[18.1%, 29.9%]	0.814

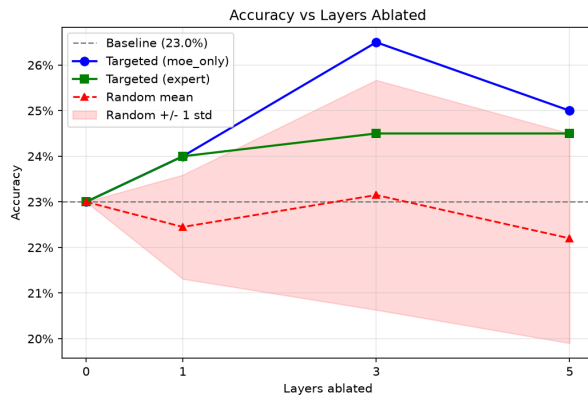
Table 6: Part A: ablation type comparison on the single worst layer (layer 41), $n = 200$ test questions. None of the three intervention types produce a statistically significant change from baseline.

MoE-only ablation performed marginally best in Part A, so we use it for Parts B and C.

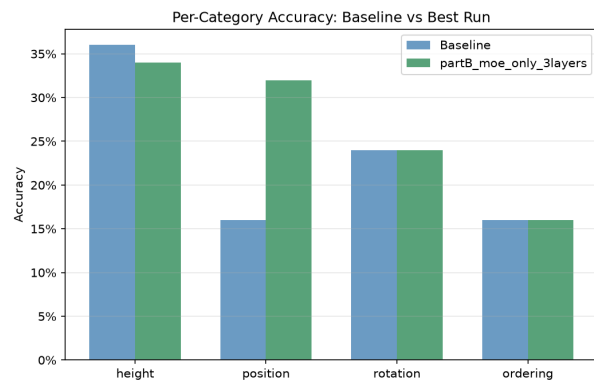
Layers ablated	Targeted accuracy	Random mean $\pm$ std	z vs. random	McNemar p (vs. baseline)
1	24.0%	$22.5\% \pm 1.1\%$	1.36	0.831
3	26.5%	$23.2\% \pm 2.5\%$	1.33	0.248
5	25.0%	$22.2\% \pm 2.3\%$	1.22	0.596

Table 7: Part B: scaling MoE-only ablation to the 1, 3, and 5 worst layers, against Part C’s random-ablation control (10 trials per layer count, mean $\pm$ standard deviation reported). z compares targeted accuracy to the random distribution; $z > 2$ is the significance threshold.

The 3-layer configuration is the best-looking result in the whole ablation study, a 3.5-point accuracy gain over baseline (23.0% to 26.5%), and the only configuration where the flipped-question count ($c = 17$ improvements, $b = 10$ regressions) clearly tilts toward improvement. Even so, McNemar’s test does not reach significance ($\chi^2 = 1.33$, $p = 0.25$), and, critically, targeted ablation of the three worst layers ($z = 1.33$) does not clear the $z > 2$ bar needed to tell it apart from simply removing three random layers.



(a) Accuracy vs. number of layers ablated, targeted vs. random.



(b) Per-category accuracy, baseline vs. best configuration (3-layer MoE-only).

Figure 6: Part B/C ablation results. The targeted curve stays within roughly one random standard deviation of the random-ablation mean at every layer count tested.

7.6 Holdout Validation

We re-ran the single best-looking configuration, MoE-only ablation of the three worst layers, on the 200-question holdout set that Steps 2 through 5 never touched. This is the decisive check: if the 3.5-point gain in **Table 7** reflects a real property of these three layers rather than noise in the 200-question test set, it should show up again, at least roughly, on new questions.

Run	Accuracy
Holdout baseline	29.0%
Holdout, 3-layer MoE-only ablation	27.0%

Table 8: Holdout validation of the best configuration (MoE-only ablation, 3 worst layers), $n = 200$ previously unseen questions.

Accuracy on the holdout set *drops* by 2 points under the same ablation that gained 3.5 points on the test set. The result does not just fail to repeat, it points the opposite way. Combined with the non-significant McNemar result and the sub-threshold random-ablation z-score in **Table 7**, we conclude the test-set gain was very likely sampling noise in a 200-question set, not evidence that layers 41, 42, and 43 are specifically responsible for the model’s spatial reasoning failures.

8 Discussion

The headline result of Phase 3 is a null result across every statistical check we ran: no ablation configuration significantly changes accuracy relative to baseline, targeted ablation never stands out from random ablation, and the one configuration that looked promising on the test set reverses direction on holdout data. We want to be precise about what this does and does not tell us.

It does not, by itself, mean that internal components are never responsible for reasoning failures in MoE vision-language models in general. It means that under this specific method, correlational z-score scoring of routing behavior, followed by Jaccard-distance clustering and gate-level or block-level ablation, we could not find such components in this model on this benchmark. Two explanations fit everything we observed, and our experiments cannot tell them apart:

1. **No localized effect exists.** Spatial reasoning failure in this model may genuinely be a distributed property of the whole network’s computation, rather than something concentrated in a few layers or experts. If so, no scoring method applied to routing statistics alone should be expected to find a strong signal, and our result is a faithful negative finding. This is plausible on independent grounds: [Elhage et al. \(2022\)](#) show that networks can represent more features than they have dedicated units for. By packing multiple sparse features into superposition within the same components, a behavior need not map onto any single expert or layer even when the underlying computation is genuinely happening somewhere in the network. Our result parallels [Wang et al. \(2026\)](#), discussed in Section 2, who find that MoE routing decisions are dominated by a small domain-invariant set of experts rather than task-specific specialists, which would independently predict exactly the kind of weak, non-generalizing per-expert signal we see.
2. **The scoring method was too weak.** Our z-scores topped out at 0.055 in magnitude (**Table 4**), far below the $z = 2$ threshold conventionally used to call a single data point an outlier, and no layer’s expert clustering reached even a modest silhouette score (**Table 5**, best case 0.236). Both point to a

weak signal-to-noise ratio in our specific measurement, not necessarily evidence that no signal exists to be found.

We cannot settle this from the data collected here, and we think that ambiguity, rather than a simple “ablation does not help,” is the more useful takeaway to report. A method that cannot tell these two explanations apart should not be used to conclude that a model’s failure mode is unfixable at the component level, it can only conclude that this particular correlational approach did not surface a fixable component. Section 10 proposes a follow-up designed specifically to tell these two explanations apart.

9 Limitations

Our sample size, 200 test questions and 200 holdout questions, is a genuine limit: per-category accuracy estimates (50 questions per category) carry confidence intervals on the order of ± 14 percentage points, and a swing of just 3 questions changes an overall accuracy estimate by 1.5 points. We tried to offset this with McNemar’s test and random-ablation controls rather than raw accuracy comparison, but a larger held-out set would sharpen every number reported here. Our z-score and clustering pipeline also only looks at the five worst-scoring layers out of 48; it is possible that a combination of moderately-scoring layers, none individually flagged, contributes more than any single layer we tested. Though the random-ablation control’s failure to tell targeted from random removal apart at any layer count argues against this being easy to find through more layer combinations either. Finally, every ablation intervention we test is all-or-nothing (an expert’s gate weight is either fully zeroed or untouched); we did not test partial suppression, which could in principle reveal graded effects that all-or-nothing ablation misses.

10 Future Work: Causal Tracing

The core limitation identified in Section 8 is that our method is correlational: it scores experts by how often they fire on wrong answers, then deletes the worst-scoring ones. It never directly tests whether a given layer’s hidden state is causally responsible for a correct answer. We propose a follow-up phase built around activation patching (Meng et al., 2022; Vig et al., 2020), a standard causal interpretability method, adapted to this benchmark as follows.

For each question, we would build a corrupted version of the input image (such as permuting image patch embeddings) strong enough to change the model’s answer, then cache the uncorrupted (“clean”) run’s hidden state after every one of the 48 decoder layers. We would then re-run the corrupted image through the model 48 times, once per layer, each time substituting in the clean run’s cached hidden state at that one layer before letting the remaining layers process as normal, and record whether the final answer flips back to the clean run’s (correct) answer. Aggregating flip rate per layer across questions gives a direct causal profile: a peaked flip-rate curve would identify the specific layer where the model resolves the visual information it needs for a correct answer, while a flat curve, unlike our current null result, would be a much stronger claim that correctness is not causally localized to any single layer’s residual stream, since it would come from a method that tests causal responsibility directly instead of through routing correlation. We see this as the natural next step for telling apart the two explanations left open in Section 8.

11 Conclusion

We tested whether specific experts or layers in a 48-layer, 128-expert Mixture-of-Experts vision-language model are responsible for its spatial reasoning failures on the LEGO-Puzzles benchmark, using z-score layer ranking, Jaccard-distance expert clustering, and a three-way ablation study validated by McNemar’s

test, a random-ablation control, and a held-out generalization check. Every statistical check returned a null result, and the one promising-looking configuration reversed direction on held-out data. We report this as a methodologically rigorous negative finding rather than a failed experiment: it rules out one specific, correlational approach to component-level localization in this model, leaves open whether a more direct causal method would find what ours could not, and proposes activation patching as that next test.

Reproducibility

All code, intermediate data, and generated figures referenced in this paper are organized by phase and step (Phase 0 through Phase 3, Steps 1 through 6 within Phase 3) in the project repository, with each step’s outputs saved before the next step consumes them, so that any stage of the pipeline can be re-run or audited independently.

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